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🧠 AutoGen Practical Quiz

🔹 Section A: Scenario-Based MCQs (Choose the BEST answer)

1. Multi-Agent Debugging System

You are building an AutoGen system where:

- Agent A: Writes Python code
- Agent B: Reviews and debugs
- Agent C: Executes code

The system is failing because Agent B is not correcting errors properly.

What is the BEST improvement?

A. Remove Agent B
B. Add prompt constraints to Agent B for structured feedback
C. Merge all agents into one
D. Increase token limit

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2. Conversational Agent Memory

A chatbot built using AutoGen forgets previous user inputs in a multi-turn conversation.

What should you implement?

A. Stateless API calls
B. Conversation memory or context window management
C. Reduce number of agents
D. Use only system prompts

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3. Agent Role Design

You created two agents:

- ResearchAgent
- SummaryAgent

Both are generating similar outputs.

What is the issue?

A. Token limit exceeded
B. Poor role separation in system prompts
C. API failure
D. Execution error

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4. Infinite Loop Problem

Two agents keep responding to each other indefinitely.

What is the BEST fix?

A. Increase temperature
B. Add termination condition or max_turns
C. Add more agents
D. Reduce prompt size

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5. Tool Usage in AutoGen

An agent needs to fetch real-time stock data.

What is the correct approach?

A. Hardcode stock values
B. Integrate tool/API calling capability
C. Use static prompts
D. Disable conversation

🔹 Section B: Code-Based Scenarios

6. Identify the Issue

```
assistant = AssistantAgent(name="assistant")
user = UserProxyAgent(name="user")

user.initiate_chat(assistant, message="Write a function to sort a list")
```

The assistant gives code but it is never executed.

What is missing?

We are missing ‘code_execution_config’ in ‘UserProxyAgent’, so the generated code is not executed.

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7. Fix the Collaboration Flow

```
agent1 = AssistantAgent(name="coder")
agent2 = AssistantAgent(name="reviewer")

agent1.initiate_chat(agent2, message="Write code for factorial")
```

Problem: Reviewer is not giving meaningful feedback.

What change would you make in agent design?

I will add a proper ‘system_message’ to the reviewer agent defining its role (e.g., code validation, feedback, etc).

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8. Execution Agent Issue

```
user_proxy = UserProxyAgent(
    name="user",
    code_execution_config=False
)
```

Why will this fail in a coding workflow?

It will fail because code_execution_config=False disables code execution, so the workflow cannot run generated code.

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9. Add a Third Agent

You have:

- Coder Agent
- Executor Agent

Now you want validation before execution.

What type of agent should you introduce and where in flow?

Validator Agent between Coder and Executor .

Flow: **Coder → Reviewer → Executor**

🔹 Section C: Design-Based Questions (Short Answers)

10. Multi-Agent Architecture Design

Design an AutoGen system for:
👉 "Automated Resume Screening"

Include:

- Agent roles
- Interaction flow

Agents and there roles:

- Resume Screener Agent → analyses resumes
- Matching Agent → compares with job requirements
- Reviewer Agent → validates shortlisted candidates

work flow:
Resume Input → Screener extracts skills → Matcher scores candidates → Reviewer validates → Final shortlisted list

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11. Conversational Agent Improvement

A chatbot gives generic answers and lacks personalization.

How would you improve it using AutoGen?

We can improve the agent by :

Add a User Profiling Agent to store preferences and context.
Use memory (chat history) + personalization prompts.

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12. Agent Collaboration Strategy

You are building a **Research Assistant** using:

- Web search
- Summarization
- Report generation

How will agents collaborate step-by-step?

Agent Collaboration Strategy will be:

Step 1: Search Agent ->collects data from web
Step 2: Summarizer-> Agent condenses information
Step 3: Research Agent ->organizes insights
Step 4: Report Generator-> creates final structured report

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13. Error Handling Scenario

During execution, one agent generates invalid Python code.

How should the system handle this collaboratively?

Executor detects error during code execution
Sends error back to Coder Agent
Coder fixes code based on feedback
Reviewer validates corrected code
Loop continues until error-free execution

🔹 Section D: Advanced Practical Scenarios

14. Real-World Workflow

Build a system for:
👉 “Automated Data Analysis from CSV”

What agents will you create and what will each do?

Agents:

- Data Loader Agent → reads CSV
- Data Analyst Agent → performs analysis (stats, trends)
- Visualization Agent → generates charts
- Insight Agent → explains results

Flow:
CSV Input → Loader → Analyst → Visualization → Insight → Final Report

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15. Prompt Engineering Challenge

An agent gives vague answers.

Rewrite the agent’s system prompt to:

- Improve clarity
- Ensure structured output

“You are an expert assistant.
Provide clear, concise, and structured responses.
Always format output as:

1. Summary
2. Key Points
3. Final Answer
   Avoid vague or generic statements.”

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16. Multi-Agent Decision System

You want agents to **vote on best solution**.

How will you design this?

We create multiple solution agents + one Judge Agent

Flow:
Problem → Multiple Agents propose solutions → Aggregator compares → selects best answer (majority vote or scoring)

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17. Scaling Issue

Your AutoGen system becomes slow with multiple agents.

What optimizations would you apply?

- Limit conversation history (use summarization)
- Reduce number of agents
- Use smaller/faster models
- Run agents in parallel where possible
- Cache repeated responses

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18. Security Scenario

An agent executes user-provided code.

What safeguards will you implement?

- Sandbox execution environment
- Restrict file/system access
- Validate and sanitize inputs
- Limit execution time/resources
- Block dangerous commands (e.g., OS/system calls)

🔹 Section E: Mini Case Study

19. End-to-End System Design

Build an AutoGen system for:

👉 **"AI Tutor for B.Tech Students"**

Requirements:

- Answer doubts
- Generate quizzes
- Evaluate answers

Design:

- Agents involved
- Collaboration flow
- Memory handling

Agents:

- Tutor Agent → explains concepts
- Quiz Generator Agent → creates questions
- Evaluator Agent → checks answers
- Feedback Agent → gives improvement tips

Flow:
Student Query → Tutor explains → Quiz generated → Student answers → Evaluator checks → Feedback provided

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20. Debug This Workflow

Flow:

1. User → Planner Agent
2. Planner → Coder
3. Coder → Executor
4. Executor → User

Problem: Output is incorrect and no validation happens.

The problem is that there is no validation step

To fix this we can add a Reviewer/Validator Agent before output

Flow:
User → Planner → Coder → Reviewer → Executor → User